A new approach to magnetopause and bow shock modeling based on automated region identification

K. Jelínek,[1] Z. Němeček,[1] and J. Šafránková[1]

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[1] The present empirical models describing a location and shape of the magnetopause and bow shock are based on a statistical evaluation of magnetopause and bow shock crossings. The crossings are usually identified by a visual inspection of the plots or by automatic methods which are less reliable. We present a new method of determination of the most probable boundary locations. The method is based on continuous plasma and magnetic field measurements in the regions visited by a sounding spacecraft (the solar wind, magnetosheath, and magnetosphere) and on the determination of ratios of these parameters to simultaneously monitored upstream parameters. The regions identified by this method are then used for development of simple models of the magnetopause and bow shock locations parameterized by the upstream pressure. The performance of the models is tested with corresponding boundary crossings based mainly on the THEMIS observations. Both developed models are in a good agreement with the results obtained from identification of crossings.

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1. Introduction

[2] The magnetopause is the obstacle varying in a size and shape in a flow of the solar wind plasma. In early published papers [e.g., *Fairfield*, 1971; *Formisano et al.*, 1973, 1979; *Sibeck et al.*, 1991; *Roelof and Sibeck*, 1993; *Petrinec and Russell*, 1996; *Shue et al.*, 1997, 1998; *Boardsen et al.*, 2000], it was found that the upstream dynamic pressure strongly influences the Earth's magnetopause position. In some of these models, a stand-off position, R is scaled with the solar wind dynamic pressure, p as $R \approx \sqrt[6]{p}$ that is based on an assumption of the dipole Earth magnetic field. On the other hand, the pressure scaling factor was included as a free fitting parameter for other models. The authors found it to be larger than 6 (e.g., 6.16 in *Boardsen et al.* [2000] or 6.6 in *Shue et al.* [1997, 1998]). By contrast, the recent papers of *Dušík et al.* [2010] and *Lin et al.* [2010] proposed the lower scaling factor of 4.8 and 5.2, respectively. The same factor (5.2) follows from an analysis of the global MHD model made by *Lu et al.* [2011].

[3] Furthermore, it was found that the second parameter driving the position and shape of the magnetopause is the B_Z component of the interplanetary magnetic field (IMF). The IMF B_Z dependence is a subject of the papers by *Sibeck et al.* [1991], *Roelof and Sibeck* [1993], *Petrinec and Russell* [1996], *Shue et al.* [1997, 1998] and many others. On the other hand, *Verigin et al.* [2009] argued that no dependence of the subsolar magnetopause position on the IMF B_Z component has been revealed in a large set of the Interball data. *Boardsen et al.* [2000] developed an empirical model of the shape of the near-Earth high-latitude magnetopause that is parameterized by the solar wind dynamic pressure, IMF B_Z component and dipole tilt angle. The authors argued that the dipole tilt angle and solar wind pressure are the most significant factors influencing the shape of the high-latitude magnetopause and that the IMF B_Z dependence can be found only if the pressure and tilt angle effects are removed by a proper scaling. Similar results were reported by *Tsyganenko* [1998] and by *Eastman et al.* [2000]. *Šafránková et al.* [2005] have analyzed high-latitude magnetopause crossings and suggested a simple correction of the *Petrinec and Russell* [1996] model that reflects the magnetopause indentation in the cusp region. This indentation was later explicitly included into the *Lin et al.* [2010] model and confirmed by statistical analysis of magnetopause and bow shock positions [e.g., *Jelínek et al.*, 2008].

[4] The solar wind plasma flow is supersonic, therefore the bow shock rises ahead of the magnetopause. The Earth's bow shock is the most studied collisionless shock [see, e.g., *Burgess*, 1995]. According to many papers, the bow shock position depends on the size and geometry of the obstacle, i.e., the magnetopause location and on the magnetosheath thickness that is a function of the Mach number. The thickness of the magnetosheath is influenced by the curvature radius, R_C of the obstacle, as it was discussed in the papers of *Farris and Russell* [1994] for the Earth's magnetosheath and *Russell and Mulligan* [2002] for the magnetosheath

[1]Faculty of Mathematics and Physics, Charles University, Prague, Czech Republic.

Corresponding Author: Z. Němeček, Faculty of Mathematics and Physics, Charles University, V Holešovičkách 2, CZ-180 00 Prague 8, Czech Republic. (zdenek.nemecek@mff.cuni.cz)

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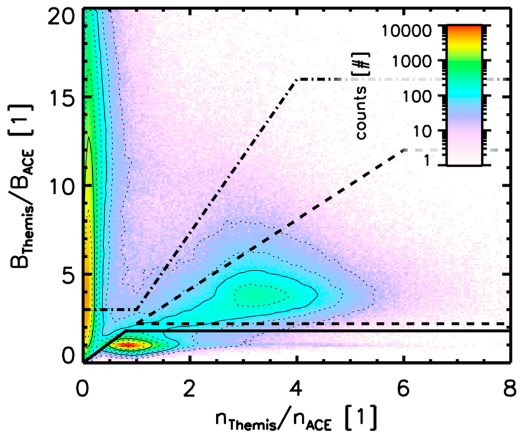

Figure 1. A 2D histogram of the ratios of r_B vs r_n which were used to distinguish three regions: the solar wind (SW) is at the bottom and it is bounded with the full line; the magnetosphere (MS) is located at the left part of the figure and bounded with the dotted-dashed line; and the magnetosheath (MSH) is a triangle distinguished with the dashed line in the middle of the panel. The values in the histogram are numbers of one-minute intervals in logarithmic scale.

of ICMEs (Interplanetary Coronal Mass Ejections). On the other hand, *Němeček and Šafránková* [1991] and *Jerab et al.* [2005] suggested that the magnetosheath thickness is a rising function of the IMF strength.

[5] The models describing actual locations and shapes of the magnetopause and bow shock as a function of upstream parameters are based on a statistical processing of crossings observed by a single spacecraft and (usually distant) solar wind monitor. This approach implicitly assumes that the downstream parameters are proportional to their upstream values. Such assumption introduces many inaccuracies when, for example, a strong sudden change of solar wind conditions results in unusual boundary crossings, or multiple crossings follow in a short time and these effects could negatively affect statistical results. Moreover, a visual inspection of data plots is time consuming and subjective because the criteria for a boundary identification could vary.

[6] Another problem is the orbital bias of particular data sets. The spacecraft on a nearly circular orbit can observe the boundary at a distance given by the orbit only, whereas the spacecraft on an elongated orbit spends a majority of time near the apogee and thus the resulting set of crossings is biased toward this apogee. This problem is important namely for the data set used in the present paper because we use the THEMIS spacecraft in the time period of 2007–2008. *Fairfield* [1971] give 14.6 R_E as the average bow shock stand-off distance and the apogee of THEMIS was 14.7 R_E in 2007. Consequently, THEMIS spent about 25% of time in distances between 13.7 and 14.7 R_E from the Earth. For this reason, *Jelínek et al.* [2009] proposed a method that partly eliminates such orbital bias. The suggested procedure weights a number of crossings observed in a particular spatial bin by a time that the spacecraft spent in this bin. However, the method cannot reflect the fact that none of crossings can be observed beyond the spacecraft apogee.

[7] For above mentioned reasons, we developed a new method of an automatic identification of both boundaries from observations of the magnetic field and plasma density. One-minute averages of these parameters measured by a sounding spacecraft are normalized to corresponding values measured at the L1 point and propagated to the sounding spacecraft location. The normalized values are then used for an identification of three regions: the solar wind, magnetosheath, and inner magnetosphere. We explain the method of data processing and apply it for the development of a simple model of the bow shock and magnetopause.

2. Data Processing

[8] For our method, we take advantage of orbits of five THEMIS spacecraft that move through all investigated regions: the solar wind (SW), the magnetosheath (MSH), and the inner magnetosphere (MS), and computed one-minute medians of the magnetic field magnitude, $|B_{Themis}|$ and density, n_{Themis}. As a solar wind monitor, we used ACE one-minute medians of the IMF magnitude, $|B_{ACE}|$, density, n_{ACE}, solar wind dynamic pressure, p, and plasma velocity, v_{ACE} shifted to THEMIS positions by convection along the X_{GSE} axis. We use two-step propagation algorithm that is described in *Šafránková et al.* [2002].

[9] For all measurements of the THEMIS spacecraft at altitudes larger than 5 R_E in the period from March 2007 to September 2009, we computed the ratio of the magnetic fields, r_B

$$r_B = \frac{|B_{Themis}|}{|B_{ACE}|}. \tag{1}$$

Because the compression ratio of the magnetic field in the magnetosheath decreases toward the flanks, we added the density compression factor, r_n

$$r_n = \frac{n_{Themis}}{n_{ACE}}. \tag{2}$$

These two ratios allowed us to identify SW, MSH, and MS regions on whole dayside parts of orbits and even toward the flanks in the range of ±7 hours of local time around the local noon.

[10] Figure 1 shows 2D histogram of r_B and r_n occurrence rates. One can clearly distinguish three regions: solar wind measurements are spread around $r_B = 1$ and $r_n = 1$ (it is not exactly the point for two reasons: shifting and comparison of distant data sources and foreshock fluctuations); the magnetosheath is specified by compression ratios of about $r_B \approx 4$ and $r_n \approx 3$ (the magnetosheath has a large spread of points around these ratios because its parameters depend on a particular position inside the magnetosheath and magnetosheath plasma and magnetic field are highly fluctuating). In many regions of the inner magnetosphere, the plasma density is small and the magnetic field does not depend on IMF and r_B can reach high values. It can be seen as a long ridge for $r_n < 1$. However, we processed all available THEMIS data regardless the spacecraft location and thus a part of magnetospheric observations was taken from the plasmasphere where the density can be higher than that in the solar wind. In Figure 1, we show the chosen boundaries

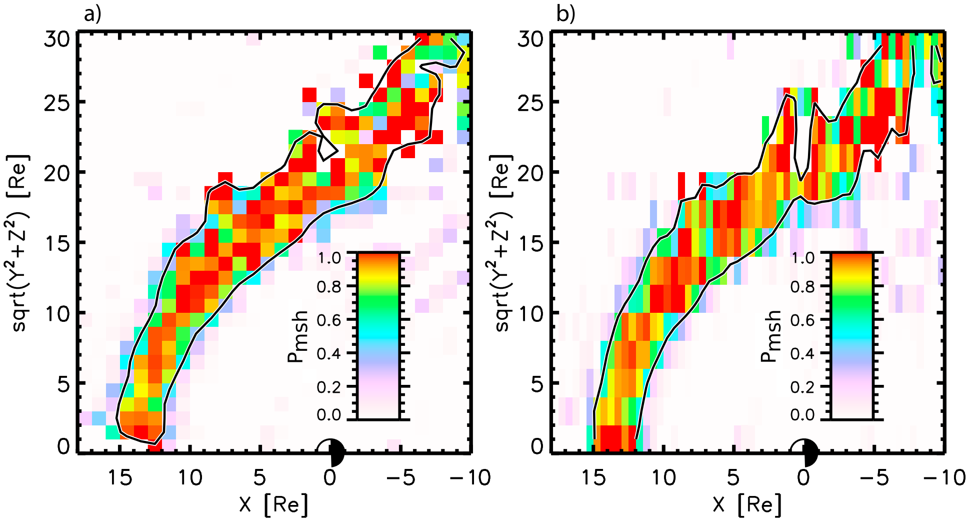

Figure 2. An example of a possible data binning in aberrated *GSE* coordinates. The color scale shows the values of the probability P_{MSH} in rectangular bins for the interval of upstream dynamic pressures from 1 to 1.1 nPa. The black line stands for an approximate location of bins with $P_{MSH} = 0.5$. (a) The 1 by 1 R_E bins. (b) The 0.5 by 2 R_E bins.

between particular regions. The points which lie outside these boundaries are not used in the further processing.

3. Coordinate System and Data Binning

[11] Since this is a first attempt to apply the above described identification of regions for a description of mean positions of the bow shock and magnetopause, we expect that their locations are controlled exclusively by the solar wind dynamic pressure. The method is based on a probability that the spacecraft at a given spatial point and under a given solar wind dynamic pressure, p visited one of regions specified in Figure 1. We assume that both the magnetopause and bow shock are rotationally symmetric around the aberrated X_{GSE} axis. The aberration takes into account the Earth orbital motion; the perpendicular components of the solar wind velocity are omitted in accord with the study of *Šafránková et al.* [2002].

[12] All data were sorted into bins described by two spatial co-ordinates X and Y and the upstream pressure, p and we computed the probability, P that we can find the solar wind (SW), magnetosheath (MSH) or magnetosphere (MS) in a particular bin.

$$P_{SW}(p, X, Y) = \frac{N_{SW}(p, X, Y)}{N_{ALL}(p, X, Y)} \quad (3)$$

$$P_{MSH}(p, X, Y) = \frac{N_{MSH}(p, X, Y)}{N_{ALL}(p, X, Y)} \quad (4)$$

$$P_{MS}(p, X, Y) = \frac{N_{MS}(p, X, Y)}{N_{ALL}(p, X, Y)}, \quad (5)$$

where N_{SW}, N_{MSH}, and N_{MS} are numbers of one-minute intervals spent in particular regions and N_{ALL} is their sum.

[13] Equations (3)–(5) do not expect any particular coordinate system. In principle, it is possible to use aberrated *GSE* coordinates and equidistant binning in space and pressure. Examples of possible binnings are shown in Figures 2a (square 1 by 1 R_E bins) and 2b (rectangular 0.5 by 2 R_E bins). The color scale shows the values of the probability P_{MSH} in aberrated *GSE* coordinates for the interval of upstream dynamic pressures from 1 to 1.1 nPa. The black line stands for an approximate location of bins with $P_{MSH} = 0.5$. We can note that the line indicates possible magnetopause and bow shock locations but the binning 1 by 1 R_E shown in Figure 2a is too coarse. On the other hand, a volume of the bin is at a lowest applicable limit because there are already several bins without measurements. The different bin shapes can improve the spatial resolution in a particular region as it is shown in Figure 2b. The bins 0.5 by 2 R_E are appropriate for the subsolar region but different shapes or orientations of the bins would be used for earlier or later local times.

[14] A possible solution is an application of non-Cartesian coordinates reflecting expected shapes of both boundaries. Since the bow shock and magnetopause surfaces are often described by second order surfaces, we expect the parabolic shapes of both boundaries. Consequently, we transform aberrated *GSE*(x, y, z) coordinates into generalized parabolic coordinates (σ, τ, ϕ) by following expressions

$$\tau = \sqrt{\sqrt{x^2 + (\lambda_y y)^2 + (\lambda_z z)^2} - x} \quad (6)$$

$$\sigma = \sqrt{\sqrt{x^2 + (\lambda_y y)^2 + (\lambda_z z)^2} + x} \quad (7)$$

$$\phi = \arctan \frac{\lambda_y y}{\lambda_z z} \quad (8)$$

where λ_y and λ_z are scaling factors.

[15] The standard parabolic coordinates can be obtained from our generalized coordinates given by equations (6)–(8) putting $\lambda_y = \lambda_z = 1$. An example of the data binning in the

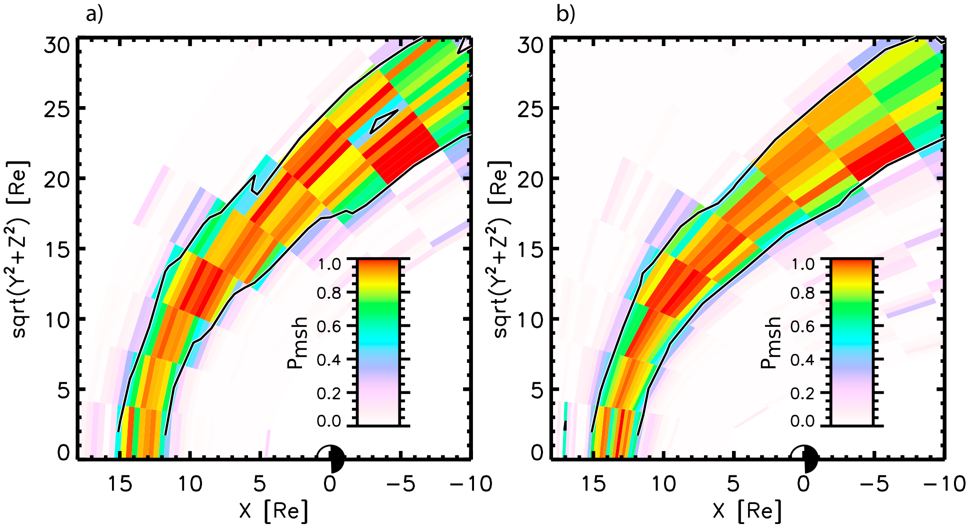

Figure 3. An example of data binning in (a) parabolic coordinates and (b) so-called magnetosheath coordinates. The definition of magnetosheath coordinates is given in the text.

standard parabolic coordinates is shown in Figure 3a and one can note that the results are better than that obtained by a binning in *GSE* coordinates shown in Figure 2. However, the matching of bins with the boundaries is not as perfect as one would expect and it is the reason why we introduced generalized coordinates that use scaling factors λ_y and λ_z. These scaling factors describe a "flaring" of parabolas in the direction of a particular axis. The bow shock or magnetopause would lie on surfaces σ = *constant* if these factors are chosen properly. Since we expect the rotational symmetry, locations of both boundaries do not depend on the angular coordinate, ϕ and we can write $\lambda_y = \lambda_z = \lambda$. However, the flaring angles of the bow shock and magnetopause surfaces are different, thus the optimum scaling factors for both boundaries would differ. Nevertheless, their values λ_{MP} and λ_{BS} can be determined by optimization procedures that are described in the next section. Expecting that the values λ_{MP} and λ_{BS} are already determined, we can make a further generalization and suppose that λ is a linear function of the distance from the Earth. The value of λ can be determined for each point A by finding the inter-sections of its radius vector, r_A with the bow shock (r_{BS}) and magnetopause (r_{MP}) surfaces from:

$$\lambda_A = \lambda_{MP} + \frac{\lambda_{BS} - \lambda_{MP}}{r_{BS} - r_{MP}}(r_A - r_{MP}) \tag{9}$$

[16] We will call these coordinates as a magnetosheath coordinate system in the further text. An example of the data binning in this coordinate system is shown in Figure 3b and we can see that it provides the best possible results for a given upstream dynamic pressure.

[17] A majority of previous studies used equidistant binning according to the solar wind dynamic pressure. However, it is well known that a response of the magnetopause or bow shock locations to the upstream pressure is non-linear. It is usually described as

$$R \approx p^{-\frac{1}{\epsilon}}. \tag{10}$$

where the suggested values of ϵ range from 4.79 [*Dušík et al.*, 2010], over 5.15 [*Lin et al.*, 2010] to 6.667 [*Shue et al.*, 1997]. To account for this non-linearity, we used the binning of the dynamic pressure as:

$$\Delta p \approx p^{+\frac{1+\epsilon}{\epsilon}}. \tag{11}$$

The value of ϵ will be specified later by the fitting to the data set.

4. The Bow Shock and Magnetopause Model

[18] There are several possibilities how to identify the bow shock and magnetopause locations from the probabilities $P(\sigma, \tau, p)$. One of them is to find locations where the probability of observations of neighboring regions (SW and MSH or MSH and MS) is equal to 0.5. Examples of this determination are shown in Figures 2 and 3. However, this way is very sensitive to missing data. For this reason, we defined an analytical expression for the bow shock and magnetopause and determined the free parameters of the model by fitting to the full data set. As already noted, we expect a parabolic shape of both boundaries that responds to the upstream pressure as $p^{-\frac{1}{\epsilon}}$. It implies that we can find the scaling factors λ_{MP} and λ_{BS} such that the bow shock or magnetopause would lie on surfaces σ = *const* in the whole range of the dynamic pressure. Using the least squares method, we have found following scaling factors: $\lambda_{BS} = 1.17$ and $\lambda_{MP} = 1.54$. In such a case, model bow shock and magnetopause positions can be expressed as

$$\sigma = \sqrt{2Rp^{-\frac{1}{\epsilon}}} \tag{12}$$

where R is the stand-off distance for p = 1 nPa and a given pressure factor, ϵ. Note that the fitting procedure uses the magnetosheath parabolic coordinates introduced above and the data are redistributed to new bins at each iteration step because our binning depends on values of λ_{MP}, λ_{BS}, and ϵ. The iteration procedure is complicated because it searches for a combination of R_{MP}, R_{BS}, ϵ_{MP}, ϵ_{BS}, λ_{MP}, and λ_{BS} such

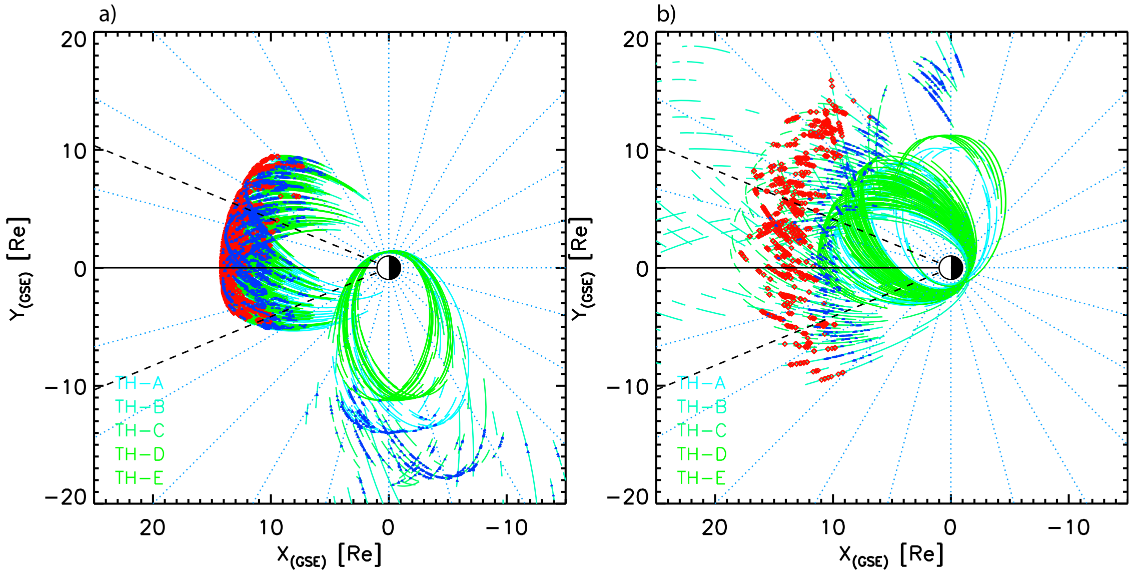

Figure 4. Distributions of bow shock (red points) and magnetopause (blue points) crossings during (a) 2007 and (b) 2008 in the X-Y plane.

that a sum of squares of distances of the bins from both boundaries given by equation (12) in the 3D space with coordinates τ, σ, p is minimized.

[19] Application of the least squares method leads to following expressions:

$$R_{MP} = 12.82\, p^{-\frac{1}{5.26}} \tag{13}$$

$$R_{BS} = 15.02\, p^{-\frac{1}{6.55}} \tag{14}$$

where R_{MP} and R_{BS}, respectively are stand-off distances of the magnetopause and bow shock, respectively. The bow shock and magnetopause surfaces in aberrated GSE coordinates can be thus written in the parametric form as

$$x = R_0\, p^{-\frac{1}{\epsilon}} - \frac{1}{2}\tau^2 \tag{15}$$

$$R_{yz} = \frac{\sqrt{2R_0\, p^{-\frac{1}{\epsilon}}\tau}}{\lambda} \tag{16}$$

where R_0 = 12.82 or 15.02 and ϵ = 5.26 or 6.55 for the magnetopause or bow shock, respectively.

5. Verification of Models

5.1. Data Preparation

[20] The described models were developed using probabilities of observations of particular regions and their performance can be tested by a comparison with real observed crossings of both boundaries. We have visually identified more than 6000 bow shock and 5500 magnetopause crossings in the THEMIS plasma and magnetic field data [*Angelopoulos*, 2008; *Auster et al.*, 2008; *McFadden et al.*, 2008] in course of the 2007 and 2008 years. These sets were biased by a low apogee of the THEMIS spacecraft, mainly during the first stage of the THEMIS project, as it can be seen in Figure 4a. Figure 4b shows distributions of magnetopause and bow shock crossings through the second stage of the THEMIS mission when the apogee reached 20 R_E. The influence of the orbital limitations is seen also in Figure 5a that shows the distribution of subsolar bow shock crossings (11–13 hours of local time) in $R - p_{dyn}$ coordinates (R stands for the distance of the bow shock from the Earth center). One can clearly see the red spot and horizontal line caused by the low apogee of THEMIS in 2007. We used the procedure of *Jelínek et al.* [2009] that weights the number of crossings by a time that the spacecraft spent in a particular bin. This method suppresses the bias only partly as Figure 5b presents because none of crossings can be observed beyond the spacecraft apogee. Nevertheless, we applied this method with a slight modification; we used parabolic coordinates and nonlinear binning of the solar wind dynamic pressure in the present paper.

5.2. Comparison of the Model and Experimental Data

[21] Figure 6 presents an example of a comparison between bow shock and magnetopause models and positions of observed bow shock (a) and magnetopause (b) crossings. The color scale shows the probability that the bow shock (magnetopause) crossing is observed in a particular spatial bin and under a given p (p is ranging from 1.0 to 1.1 nPa in the figure). One can note a good matching of crossings and model results. Histograms in Figure 7 represent the differences between average positions of the bow shock (a) and the magnetopause (b) determined from boundary crossings and from the proposed models. Both distributions are almost centered.

[22] Since the empirical models of boundaries are usually developed by fitting to observed crossings, we can follow this approach and test our model by this way. We divided the probability of crossing observations (see Figure 5b) into bins in parabolic coordinates and applied the above described fitting procedure to these sets of probabilities. The resulting

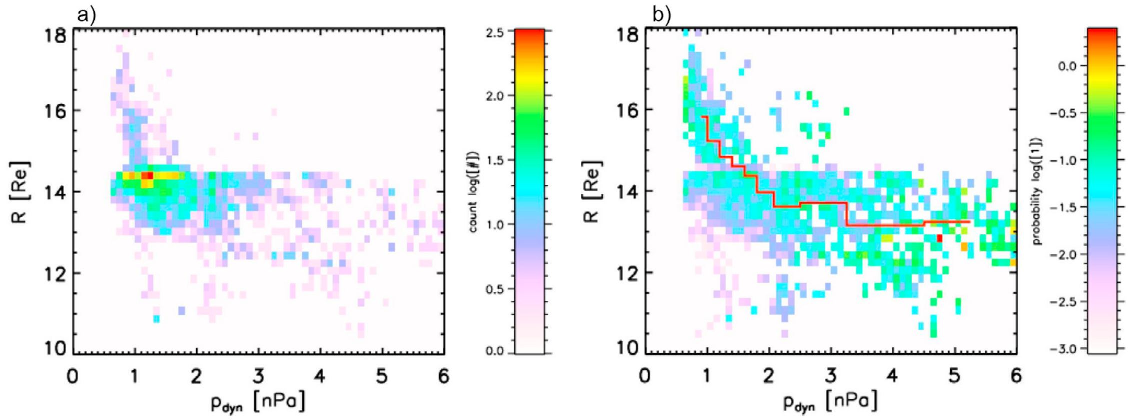

Figure 5. (a) The distribution of subsolar bow shock crossings (11–13 hours of local time) in $R - p_{dyn}$ coordinates when R stands for the distance of the bow shock from the Earth center and p_{dyn} is the dynamic pressure. (b) The same histogram weighed by a time that the spacecraft spent in a particular bin. The red line represents average bow shock location.

stand-off positions of the magnetopause, R_{MP} and bow shock, R_{BS} can be written in a similar form to equations (13) and (14)

$$R_{MP} = 12.90\ p^{-\frac{1}{4.92}} \quad (17)$$

$$R_{BS} = 14.94\ p^{-\frac{1}{6.62}}. \quad (18)$$

Comparing both sets of the equations (13)–(14) and (17)–(18), we can conclude that values R_0 and ϵ are close to the suggested models. This result validates our method but, as mentioned above, the sets of crossings suffer with the orbital bias and it is not clear if the method of *Jelínek et al.* [2009] can remove this bias completely. On the other hand, the method used for the model development in the present paper does not depend on the spacecraft orbits because it searches for a place where the probability of observations of two neighboring regions is equal to 0.5. If the spacecraft apogee is too low, such place simply would not be found.

6. Discussion

[23] The described method relies on the identification of the regions and we use the magnetic field and ion density for this purpose. The identification could be more reliable if the number of used parameters would be extended. We tried to add the ratio of velocities, however, it reduced the data set to approximately one quarter due to missing data and did not bring any significant improvement of the region identification. Nevertheless, we believe that it is the right way for a further enhancement of our method.

[24] A surprisingly low value of ϵ in the case of the magnetopause can be probably explained by a pressure of plasma

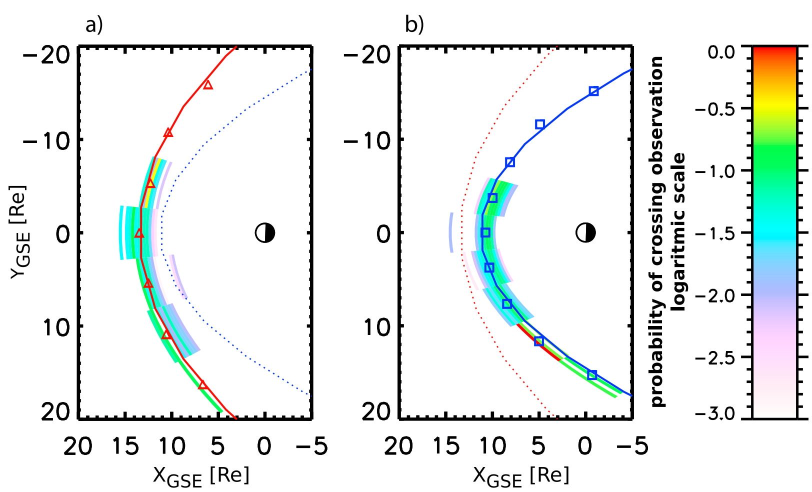

Figure 6. A comparison of the presented (a) bow shock and (b) magnetopause models with the observed bow shock and magnetopause crossings. The color scale shows the probability of observation of the particular boundary crossing in a given bin. The crossings from the pressure range of 1.0–1.1 nPa are shown and the full line stands for the model boundary location under 1.05 nPa of the dynamic pressure (Figure 6a – red line, Figure 6b – blue line). The second boundary (Figure 6a – the magnetopause, Figure 6b – the bow shock) is given by the dotted line for the sake of reference.

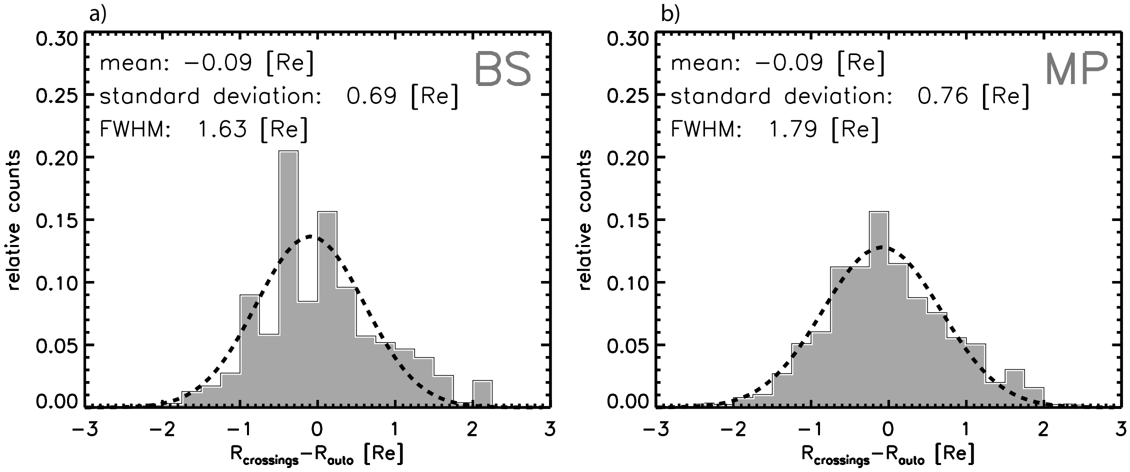

Figure 7. (a) Distributions of differences between positions of observed bow shock crossings and model positions. (b) The same plot for the magnetopause. The parameters of the Gaussian fit are given at the top of each panel.

inside the magnetosphere. However, we should note that the recent papers of *Lin et al.* [2010] and *Dušík et al.* [2010] report even lower exponents, ≈5.15 and 4.79, respectively. We have tested the dependence of the magnetopause location on the IMF B_Z sign but we did not find any statistically significant difference between sets for large positive or negative IMF B_Z. We will return to this point after collection of a significantly larger data set.

[25] The expression for the bow shock does not take into account the Mach number, however, the dependence of the bow shock location on the Mach number is expected to be rather weak for $M > 4$ and this condition is fulfilled for a majority of the observations. On the other hand, different values of ϵ for the bow shock and magnetopause suggest that the magnetosheath thickness would be a slightly rising function of the solar wind dynamic pressure. Since both the Mach number and solar wind dynamic pressure depend on the upstream density and speed, the change of the magnetosheath thickness with solar wind dynamic pressure following from equations (13)–(14) would be apparent and connected with well established dependence of this thickness on the upstream Mach number.

[26] The width of the distribution in Figure 7 seems to be large (FWHM ≈ 1.7 R_E) but we can compare this width with other models. The analysis of *Šafránková et al.* [2002] provides FWHM varying from ≈1.2 to ≈1.9 R_E for different magnetopause models and the value 1.7 R_E lies in this interval. Moreover, we applied our model on the set ≈1700 of dayside magnetopause crossings described in *Šafránková et al.* [2002] and found that FWHM = 1.3. Figure 8 presents the histogram of relative deviations of modeled and observed crossings, R_{MOD}/R_{OBS} in the same form as the histograms in the *Šafránková et al.* [2002] paper for an easier comparison. This result is fully comparable with another empirical models [*Petrinec and Russell*, 1996; *Shue et al.*, 1997, 1998] in the subsolar region at low latitudes.

[27] *Lin et al.* [2010] developed a very complex magnetopause model parameterized with the dynamic pressure, IMF B_Z, and tilt angle. They tested their model as well as several older models using a set of 62 observed low-latitude crossings and found that the values of standard deviations are in the range 0.65–1 R_E and only their model provided standard deviations of 0.54 R_E. From this comparison it follows that our result (0.76 R_E) is relatively good if we take into account that we incorporate only the dependence on the solar wind dynamic pressure.

[28] However, our model is based on THEMIS observations that cover only very limited ranges of latitudes and tilt angles. In spite of its relatively good performance, the model cannot describe the dependence of the magnetopause location on the tilt angle [*Boardsen et al.*, 2000; *Šafránková et al.*, 2005; *Jelínek et al.*, 2008; *Lin et al.*, 2010] or elliptical magnetopause cross-section suggested by, e.g., *Roelof and Sibeck* [1993], *Boardsen et al.* [2000], and *Lin et al.* [2010]. Such effects could be included after enlargement of the original data set with observations of other spacecraft like Cluster or earlier Prognoz type satellites that map the high-latitude regions. With such data set, we could account for different curvature radii of the magnetopause nose in the X-Y and X-Z planes simply assuming that λ_y and λ_z in equations (6)–(8) are not equal and determining their values

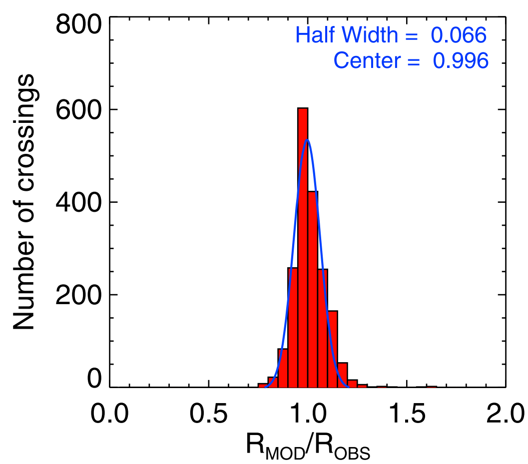

Figure 8. Histogram of R_{MOD}/R_{OBS} ratios for the present magnetopause model and the set of crossings in *Šafránková et al.* [2002].

by the fitting procedure. Moreover, since Cluster and THEMIS operate in the dayside regions in different parts of year, the combined data set can cover the full range of tilt angles, thus the dependence of model parameters on the tilt angle could be resolved.

[29] Nevertheless, our model(s) can serve as a very useful tool for an analysis of any data from the region covered by the measurements used for its(their) development. An example of their application can be found in the study of pressure profiles in the magnetosheath by *Samsonov et al.* [2012].

7. Conclusion

[30] We have developed a new automated method for identifications of bow shock and magnetopause positions and shapes. We successfully validated this method using ≈6000 BS and ≈5500 magnetopause dayside crossings that we have identified by a visual inspection of THEMIS data plots. Taking into account the limitations of the used data set, we can recommend the application of our models for the dayside low-latitude region in the range of solar wind dynamic pressures from 0.6 to 11 nPa.

[31] The main advantage of the suggested method is that it can be used for development of the magnetopause and bow shock models without necessity to identify boundary crossings. We have applied the method on THEMIS observations but it can be used for any spacecraft (or ensemble of the spacecraft including those on a geostationary orbit) orbiting in the near-Earth space and providing the plasma density and magnetic field strength if the simultaneous solar wind and IMF monitoring is available.

[32] Both models are in a good agreement with results obtained from observed boundary crossings and they are comparable with the previous more complex models. We plan to significantly enlarge the input data set and to extend the number of control parameters, especially to add IMF B_Z and the tilt angle for the magnetopause, and the Mach number for the bow shock.

[33] Another direction that can enhance the results of the bow shock and magnetopause modeling is an application of different non-Cartesian coordinate systems for the magnetopause and bow shock. A simple comparison of models with observed crossings in different local times (not shown) revealed that whereas the generalized parabolic coordinates provide good results for the bow shock, a modification of elliptical coordinates would be more appropriate for the description of the magnetopause location.

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